

## Decimal calculations



1 Use a calculator to evaluate the following:

a  $0.96 \times 0.76$

b  $7.4 \times 0.065$

c  $4780 \times 3.58 \times 0.922$

d  $0.086 \div 8.4$

e  $1.36 \times 0.65 \div 2$

f  $83 \times (0.12 + 0.013)$

g  $12.4 \div 0.58 \div 0.844$

h  $52.3 \div (0.47 \times 78)$

2 Use a calculator to convert the following fractions into decimals, rounded to three decimal places:

a  $\frac{3}{0.29}$

b  $\frac{8.79}{40.5}$

c  $\frac{11}{125}$

d  $\frac{31}{16}$

e  $\frac{379}{512}$

f  $\frac{22.5}{8.8}$

g  $\frac{2.6}{65}$

h  $\frac{4.62}{15}$

## Percentage calculations

3 Use a calculator to express the following as percentages:

a  $\frac{15}{8}$

b  $\frac{581}{700}$

c  $\frac{291}{485}$

d  $\frac{357}{680}$

e 44 out of 64

f 899 out of 725

g 5.9 out of 0.32

h 76 out of 91.2

4 Use a calculator to convert the following into decimals:

a  $\frac{33\,100}{25}\%$

b 9.25%

c  $51\% \times 83\%$

d 70% of 617

e  $135\% \times 24.6$

f  $\frac{67.2}{24}$

g  $\frac{3}{4}$  of 65.86

h  $72\% \div 3$

5 Use a calculator to find the following:

a 23% of  $\frac{54}{17}$

b 91.6% of 97.3

c 317% of  $\frac{23.6}{8}$

d 126% of 87.2

e 209% of  $\frac{29}{22}$

f 47.3% of 1.764

6 Use a calculator to determine  $x$  in each of the following:

a 70% of  $x$  is equal to 42.7

b 60% of  $x$  is equal to 39.54

c 60.1% of  $x$  is equal to 24.04

d 46% of  $x$  is equal to 35.88

- 7 Give some examples of when it might be better or not to use a calculator in percentage calculations.



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## Applications

- 8 The 64 girls at Victoria's party represent 80% of all the guests. Find the total number of guests at the party.
- 9 A city has a yearly water supply of 27 000 000 megalitres. It gets 14% of this water supply from melted snow. Find the amount of water that comes from melted snow each year.
- 10 A bag of potatoes weighs 50 kg. When more potatoes are added, the weight of the bag increases by 35% find the new weight of the bag of potatoes.
- 11 Duncan is going to buy a hat that is marked as 25% off. The original price was \$30.
- a Find the discount amount.
  - b Find the discounted price that Duncan will pay for the hat.
- 12 Ryan scored 60% on her Maths exam, which was marked out of 110. What was Ryan's actual mark out of 110?
- 13 Jane paid \$750 to get a dress made. \$450 of this was for the materials and the rest was for the labour required.
- a Find the percentage of the cost that paid for the materials.
  - b Find the percentage of the cost that paid for the labour.